

Discrete Structure for Computing. Assignment #3

1. Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach

b) $a_n = a_{n-1} + 3, a_0 = 1$

c) $a_n = a_{n-1} - n, a_0 = 4$

d) $a_n = 2a_{n-1} - 3, a_0 = -1$

e) $a_n = (n + 1)a_{n-1}, a_0 = 2$

2. Find $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if f is defined recursively by $f(0) = f(1) = 1$ and for $n = 1, 2, \dots$

a) $f(n + 1) = f(n) - f(n - 1)$.

b) $f(n + 1) = f(n)f(n - 1)$.

c) $f(n + 1) = f(n)2 + f(n - 1)3$.

d) $f(n + 1) = f(n)/f(n - 1)$.

3. A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

4. Use the Euclidean algorithm to find

a) $\gcd(12, 18)$. b) $\gcd(111, 201)$.

c) $\gcd(1001, 1331)$.

d) $\gcd(12345, 54321)$.

e) $\gcd(1000, 5040)$.

f) $\gcd(9888, 6060)$.

5. Show that if n is a positive integer, then $1 + 2 + \dots + n = n(n + 1) / 2$

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

6. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .

- a) What is the statement $P(1)$?
- b) Show that $P(1)$ is true, completing the basis step of the proof.
- c) What is the inductive hypothesis?
- d) What do you need to prove in the inductive step?
- e) Complete the inductive step, identifying where you use the inductive hypothesis.

7. Let p and q be the propositions “The election is decided” and “The votes have been counted,” respectively. Express each of these compound propositions as an English sentence.

- a) $\neg p$ b) $p \vee q$ c) $\neg p \wedge q$
- d) $q \rightarrow p$
- e) $\neg q \rightarrow \neg p$ f) $\neg p \rightarrow \neg q$
- g) $p \leftrightarrow q$ h) $\neg q \vee (\neg p \wedge q)$

8. Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express each of these propositions as an English sentence.

- a) $p \rightarrow q$
- b) $\neg q \leftrightarrow r$
- c) $q \rightarrow \neg r$
- d) $p \vee q \vee r$
- e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
- f) $(p \wedge q) \vee (\neg q \wedge r)$

9. Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations)

a) You get an A in this class, but you do not do every exercise in this book.

b) You get an A on the final, you do every exercise in this book, and you get an A in this class.

c) To get an A in this class, it is necessary for you to get an A on the final.

d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.

e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

10. Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

11. Suppose that f is defined recursively by $f(0) = 3$, $f(n + 1) = 2f(n) + 3$. Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$

Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1}$$

$$b_n = a_{n-1} + 2b_{n-1}$$

with $a_0 = 1$ and $b_0 = 2$.

12. In a discrete mathematics class, every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

13. Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

14. A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

15. There are 345 students at a college who have taken a course in calculus, 212 who have taken a course in discrete mathematics, and 188 who have taken courses in both calculus and discrete mathematics. How many students have taken a course in either calculus or discrete mathematics?